

SECTION A – MCQs with Explanations (20 × 3 = 60 marks)

Q1. Which of the following best captures the philosophy behind MLOps?

- A. Manual deployment of ML models by data scientists
- B. Automation and monitoring of the entire ML lifecycle
- C. Only version controlling code changes
- D. Scaling only the data storage layer

Answer: B. MLOps automates and monitors the entire lifecycle—from data ingestion to deployment and monitoring.

A is wrong because MLOps removes manual work; *C* and *D* address only parts of the system, not the full lifecycle.

Q2. What is the main reason models fail in production?

- A. Incorrect hyperparameters
- B. Poor container orchestration
- C. Train-serve skew and data drift
- D. Slow CI/CD pipelines

Answer: C. When live (serving) data differs from training data, accuracy drops silently.

A and *D* affect efficiency, not accuracy; *B* affects uptime, not prediction correctness.

Q3. The command to create a Conda environment with Python 3.8 is:

- A. `python -m venv env38`
- B. `conda install python=3.8`
- C. `conda create --name env38 python=3.8`
- D. `conda activate env38 python=3.8`

Answer: C. The `create` keyword makes a new environment.

A uses `venv`, not Conda; *B* installs Python globally; *D* activates an already-existing env.

Q4. Which of the following commands exports all packages of a Conda environment to YAML?

- A. `conda export > env.yml`
- B. `conda env export > environment.yml`
- C. `conda list --export`
- D. `conda freeze > requirements.txt`

Answer: B. Correct command for Conda YAML export.

A invalid syntax; *C* lists but doesn't export YAML; *D* is for pip, not Conda.

Q5. Which layer in a Docker image defines the base OS?

- A. RUN
- B. COPY
- C. FROM
- D. ENTRYPOINT

Answer: C. **FROM** sets the base image layer (e.g., **FROM centos:8**).

A adds software; *B* adds files; *D* defines the default process.

Q6. Consider the Dockerfile line **ENTRYPOINT ["/bin/bash"]**. What does it do?

- A. Runs bash every time the container starts
- B. Builds bash into the image
- C. Installs bash via yum
- D. Deletes previous ENTRYPOINT commands

Answer: A. ENTRYPOINT defines the default process.

B and *C* refer to build-time, not run-time; *D* is invalid.

Q7. Which statement about containers is *true*?

- A. Containers virtualize hardware.
- B. Containers share the host kernel.
- C. Containers boot slower than VMs.
- D. Containers can't run multiple OS types.

Answer: B. Containers share the host kernel for lightweight isolation.

A describes VMs; *C* false—they start in seconds; *D* false—you can emulate OS types.

Q8. The component managing multiple containers across nodes is called:

- A. Dockerfile
- B. Hypervisor
- C. Kubernetes
- D. GitHub Actions

Answer: C. Kubernetes orchestrates containers across clusters.

A defines image; *B* for VMs; *D* for CI/CD, not orchestration.

Q9. Which scaling approach adds *more machines* rather than resources to one?

- A. Vertical scaling
- B. Horizontal scaling
- C. Diagonal scaling
- D. Resource scaling

Answer: B. Horizontal scaling = adding nodes/pods.

A upgrades CPU/RAM; C is not standard; D vague.

Q10. What command builds a Docker image tagged “mlapp:v1”?

- A. `docker run mlapp:v1 .`
- B. `docker build -t mlapp:v1 .`
- C. `docker push mlapp:v1`
- D. `docker image create mlapp:v1 .`

Answer: B. Correct syntax for image build.

A runs, not builds; C pushes to registry; D invalid.

Q11. The key function of a **Feature Store** is to:

- A. Visualize model predictions
- B. Store and reuse curated ML features
- C. Replace a data warehouse
- D. Handle inference latency

Answer: B. Feature Stores manage reusable features and reduce train-serve skew.

A visualization is BI; C data warehouses serve BI, not ML; D only indirectly related.

Q12. What differentiates AutoML tools like Ludwig and FLAML?

- A. They only deploy models.
- B. They automate training, selection, and tuning.
- C. They require manual feature selection.
- D. They can't log experiments.

Answer: B. AutoML automates data prep, feature engineering, and tuning.

A too narrow; C opposite; D false—many log with MLflow.

Q13. MLflow's `start_run()` function is used to:

- A. Start a Docker container
- B. Begin a new experiment tracking session
- C. Push model to registry
- D. Log metrics to Prometheus

Answer: B. Starts a new MLflow tracking run to log parameters/metrics.

A and C different systems; D unrelated tool.

Q14. Which cloud service provides fully managed Kubernetes?

- A. AWS ECS
- B. Azure Batch AI
- C. AWS EKS
- D. GCP Pub/Sub

Answer: C. EKS is AWS's managed Kubernetes.

A is Docker-only; B for training, not orchestration; D is a messaging service.

Q15. "Concept drift" refers to:

- A. Change in input data distribution
- B. Change in relationship between features and target
- C. Incorrect container configuration
- D. Loss of model weights

Answer: B. Concept drift = target concept shifts; data drift = feature distribution change.

A is data drift; C and D irrelevant.

Q16. Which best defines a microservice?

- A. Large monolithic architecture
- B. Self-contained function with minimal dependencies
- C. VM hosting multiple models
- D. Docker container registry

Answer: B. Each microservice performs one independent function, e.g., ML prediction API.

Q17. The Git command to create and switch to a new branch is:

- A. `git switch branchname`
- B. `git branch branchname && git checkout branchname`

- C. `git checkout -b branchname`
- D. `git add branchname`

Answer: C. Combines creation + switch.
A incomplete; B two-step older style; D invalid.

Q18. In CI/CD pipelines, the term “Quality Gate” means:

- A. Git branch name
- B. Check that blocks merges if performance drops
- C. Cloud firewall policy
- D. Docker network restriction

Answer: B. Used in LLMOps to stop bad model/prompts from deploying.

Q19. In cloud MLOps, which pairing is correct?

- A. Azure → SageMaker B. GCP → Vertex AI
- C. AWS → Synapse D. Azure → Bedrock

Answer: B. GCP’s MLOps suite = Vertex AI.
A AWS tool mismatch; C Synapse is Azure; D Bedrock is AWS LLM suite.

Q20. The command to push a local Docker image to Docker Hub is:

- A. `docker upload user/mlapp:v1`
- B. `docker push user/mlapp:v1`
- C. `docker run user/mlapp:v1`
- D. `docker image hub user/mlapp:v1`

Answer: B. Push sends image to registry. Others are syntactically invalid.

SECTION B – Short Answer Questions (10 × 4 = 40 marks)

Q1. Explain the *MLOps lifecycle* with all key stages.

Answer:

MLOps extends DevOps to the ML domain, covering data management, model experimentation, versioning, deployment, and monitoring. Data pipelines ensure consistent input, while experiment tracking (e.g., MLflow) logs metrics. Model registries maintain version

control, and CI/CD automates retraining and deployment. Continuous monitoring detects drift and performance degradation, closing the feedback loop.

Q2. Compare `Virtualenv` and `Conda` environments.

`Virtualenv` is Python-native, lightweight, and pip-based, suitable for isolated Python projects. `Conda`, from Anaconda, supports both Python and non-Python binaries, ideal for data science environments. `Virtualenv` stores dependencies in the project folder; `Conda` keeps them under `/envs/`. `Conda` environments can be exported via `conda env export > environment.yml` and recreated on any system, ensuring reproducibility across teams.

Q3. Differentiate `Containers` and `Virtual Machines` in terms of architecture, performance, and use cases.

VMs virtualize hardware with a hypervisor, giving full OS isolation but high overhead. Containers virtualize only the OS layer, sharing the kernel with the host, making them lightweight and faster to boot. Containers are better for microservices and cloud-native apps, while VMs suit security-critical workloads.

Q4. Write a simple `Dockerfile` to run Python 3.8 on CentOS with bash entrypoint and explain each line.

```
FROM centos:8          # Base layer
RUN dnf install -y python38 # Install Python
ENTRYPOINT ["/bin/bash"]  # Default shell when container runs
```

`FROM` defines base OS, `RUN` executes commands during build (creates a new layer), and `ENTRYPOINT` specifies default process.

Q5. What is `Horizontal vs Vertical Scaling` in Kubernetes?

Vertical scaling increases resources (CPU/RAM) of a single node. Horizontal scaling adds more pods/nodes to distribute load—handled by the Kubernetes Horizontal Pod Autoscaler (HPA). Horizontal scaling improves fault tolerance and elasticity, critical for real-time ML inference services.

Q6. Explain how a `Feature Store` resolves train-serve skew.

A Feature Store centralizes precomputed, versioned features shared between training and

serving. It ensures the same transformation logic applies in both phases. This prevents mismatched data pipelines (e.g., customer age calculated differently). Online stores serve low-latency predictions; offline stores feed training pipelines.

Q7. What is **Drift Detection** and how is it monitored in MLOps?

Drift detection identifies when live data or model behavior deviates from training baselines.

Data drift means feature distribution shifts; **concept drift** means target relationships change.

Tools like **Evidently AI** or **SageMaker Model Monitor** automate drift alerts, enabling retraining triggers to maintain reliability.

Q8. Describe **MLflow tracking** and how to log experiments.

MLflow tracks experiments by connecting to a tracking server (URI). Each run

(`mlflow.start_run()`) logs parameters (`log_param`), metrics (`log_metric`), and artifacts (models or plots). This enables versioned comparisons and reproducible experiments.

Integrating it with CI/CD ensures automated evaluation of new model versions.

Q9. How do **CI/CD pipelines** integrate with ML model deployment?

CI/CD automates building, testing, and deploying ML containers. Continuous Integration (CI) runs unit tests and builds images; Continuous Delivery (CD) pushes to staging or production (via GitHub Actions, Jenkins). In MLOps, it extends to data validation, retraining, and model registry promotion, forming an automated loop.

Q10. Compare **AWS SageMaker**, **Azure ML**, and **GCP Vertex AI** for MLOps capabilities.

All three provide managed compute, data pipelines, feature stores, experiment tracking, and endpoints. AWS SageMaker is end-to-end with JumpStart and Bedrock for LLMs; Azure ML integrates with Prompt Flow and AKS; Vertex AI unifies training, tuning (Vizier), and monitoring under one API. Each offers cost and latency tracking for observability.